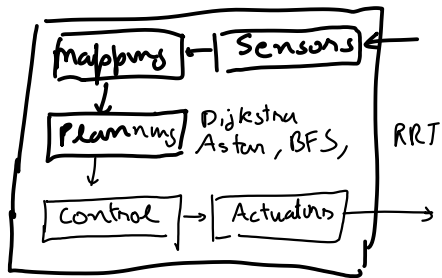


Control

Mapping



Robotic arm
 $R, T \begin{cases} \text{Forward Kinematics (FK)} \\ \text{Inverse Kinematics (IK)} \end{cases}$

Planning : Going from start state to goal state

Higher-level of abstraction

Long trajectories

Trajectory in terms of states



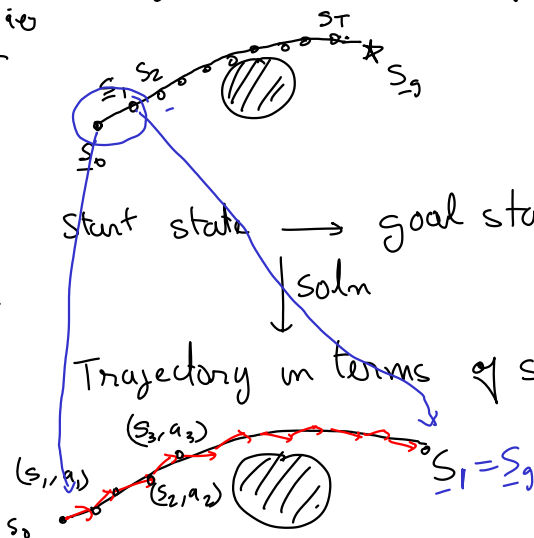
MEE 444
 Robotic control

Control : Start state $\rightarrow$ goal state

Lower level of abstraction

Trajectory in terms of state-action pairs / policy function
 state feedback control

Follow the trajectory provided by the Planner



Control

- PID : Proportional - Integral - Differential controller $\begin{pmatrix} P \\ PI \\ PD \end{pmatrix}$
- LQR : Linear Quadratic Regulator $\rightarrow$ iLQR incremental LQR
- MPC : Model Predictive control
- RL : Reinforcement Learning $\rightarrow$ Q-learning (Tabular form)

PID controller

single dimensional control

for multi dimensional control use one PID per dimension

For Jetbot: what is the state space, action space, system dynamics, and cost function?

We only covered Discrete planning

But Jetbot has states and action in cont. domain.

Simplified mathematical models for robot like Jetbots

↳ Unicycle model ✓

↳ Ackermann drive



$$\text{state} = \begin{bmatrix} x \\ y \\ \theta \end{bmatrix} = \underline{s}_t$$

$\underline{s}_t \in \mathcal{S}$
 $\uparrow$
 all possible
 values of
 state
 = State space

$$\underline{a}_t = \text{action} = \begin{bmatrix} v_t \\ \omega_t \end{bmatrix}$$

$\uparrow$
 control
 signal (u_t)

v_t ← linear velocity
 ω_t ← angular velocity

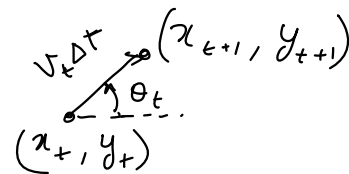
$\underline{a}_t \in \mathcal{A}$
 $\uparrow$
 Action
 space

$\underline{u}_t$ RL
 $\underline{a}_t$

System dynamics (discrete time, Δt)

$$\underline{s}_{t+1} = f(\underline{s}_t, \underline{a}_t)$$

$$\underbrace{\begin{bmatrix} x_{t+1} \\ y_{t+1} \\ \theta_{t+1} \end{bmatrix}}_{\underline{s}_{t+1}} = \underbrace{\begin{bmatrix} x_t + v_t \Delta t \cos \theta_t \\ y_t + v_t \Delta t \sin \theta_t \\ \theta_t + \omega_t \Delta t \end{bmatrix}}_{f(\underline{s}_t, \underline{a}_t)}$$



Acceleration

Markovian
state

must capture
all the info

$$\underline{s}_t = \begin{bmatrix} x_t \\ y_t \\ v_t \\ \theta_t \end{bmatrix}$$

$$\underline{a}_t = \begin{bmatrix} b_t \\ \omega_t \end{bmatrix}$$

b_t ← linear
acc
 ω_t ← ang.
vel

$$\underline{s}_{t+1} \perp \underline{s}_{t-1} \mid \underline{s}_t$$

$\underline{s}_{t+1}$ is conditionally ind. of $\underline{s}_{t-1}$
given $\underline{s}_t$

that is required to predict next
state

cost function = defines the task

$$c(s_t, a_t) = \|s_t - s_g\|^2 + \lambda \|a_t\|^2 \quad \leftarrow \text{example}$$

Control problem

$$\{a_t\}_{t=1}^T = \underset{\{a_t\}_{t=1}^T}{\text{minimize}} \quad \sum_{t=1}^T c(s_t, a_t)$$

$$\text{subject to} \quad s_{t+1} = f(s_t, a_t) \\ \forall t \in \{1, \dots, T\}$$

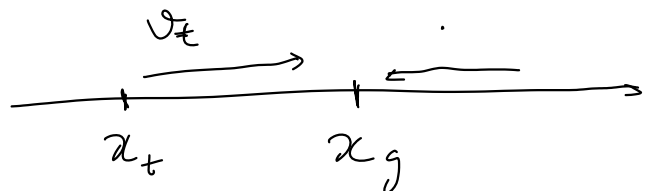
policy function $\uparrow$

$$\pi^*(s_t) = \underset{\pi}{\text{minimize}} \quad \sum_{k=t}^T c(s_k, \pi(s_k))$$
$$\text{subject to} \quad s_{k+1} = f(s_k, \pi(s_k)) \\ \forall k \in \{t, \dots, T\}$$

PID controller does not use system dynamics or the cost function. Hence it is not called optimal control

We construct an error function (must be signed)

$$e(x_t) = x_g - x_t$$



$$a_t = \underbrace{k_p e(x_t)}_{\text{Proportional term}} + \underbrace{k_I \int_0^t e(x_t) dt}_{\text{Integral}} + \underbrace{k_d \cdot \frac{d}{dt} e(x_t)}_{\text{Differential term}}$$

Linear Quadratic Control/Regulator (LQR)

$$\{a_t\}_{t=1}^T = \arg \min_{\{a_t\}} \sum \text{cost function} \quad \begin{array}{l} \text{Quadratic} \\ \text{in state} \\ \text{and action} \end{array}$$

subject to system dynamics
Linear in state and action

$$c(\underline{s}_t, \underline{a}_t) = \underline{s}_t^T Q_t \underline{s}_t + \underline{a}_t^T R_t \underline{a}_t + \underline{s}_t^T N_t \underline{a}_t + \underline{q}_t^T \underline{s}_t + \underline{r}_t^T \underline{a}_t$$

Using homogeneous coordinates linear terms like $\underline{q}_t^T \underline{s}_t$ can be absorbed in the quadratic term.

$$\tilde{\underline{s}}_t = \begin{bmatrix} \underline{s}_t \\ 1 \end{bmatrix} \Rightarrow \underline{s}_t^T Q_t \underline{s}_t + \underline{q}_t^T \underline{s}_t \stackrel{?}{=} \tilde{\underline{s}}_t^T \tilde{Q}_t \tilde{\underline{s}}_t$$

$$= \begin{bmatrix} \underline{s}_t^T & 1 \end{bmatrix} \begin{bmatrix} \underline{Q} & \underline{x} \\ \underline{x}^T & 1 \end{bmatrix} \begin{bmatrix} \underline{s}_t \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} \underline{s}_t^T & 1 \end{bmatrix} \begin{bmatrix} \underline{Q} \underline{s}_t + \underline{x} \\ \underline{x}^T \underline{s}_t + 1 \end{bmatrix} \rightarrow \underline{s}_t^T \underline{Q} \underline{s}_t + \underline{s}_t^T \underline{x} + \underline{x}^T \underline{s}_t + 1 \quad \in \mathbb{R}$$

$$= \underline{s}_t^T \underline{Q} \underline{s}_t + \underline{s}_t^T \underline{x} + \underline{x}^T \underline{s}_t + 1$$

$$(\underline{s}_t^T \underline{x})^T = \underline{x}^T \underline{s}_t = \underline{x}^T \underline{s}_t$$

$$\underline{s}_t^T Q \underline{s}_t + \underline{q}^T \underline{s}_t = \underline{s}_t^T Q \underline{s}_t + 2 \underline{x}^T \underline{s}_t + 1$$

$$\Rightarrow 2 \underline{x} = \underline{q} \Rightarrow \underline{x} = \underline{q}/2$$

$$\underbrace{\underline{s}_t^T Q \underline{s}_t}_{\text{quad}} + \underbrace{\underline{q}^T \underline{s}_t}_{\text{linear}} = \underbrace{\tilde{\underline{s}}_t^T \hat{Q} \tilde{\underline{s}}_t}_{\text{quad}} - 1$$

homogeneous coordinates

$$\hat{Q} = \begin{bmatrix} Q & \underline{q}/2 \\ \underline{q}^T/2 & 1 \end{bmatrix} \quad \tilde{\underline{s}}_t = \begin{bmatrix} \underline{s}_t \\ 1 \end{bmatrix}$$

LQR:

$$\text{minimize } \sum_{t=1}^T \underline{s}_t^T Q \underline{s}_t + \underline{a}_t^T R \underline{a}_t$$

{ $\underline{a}_t$ }_{t=1}^T

subject to $\underline{s}_{t+1} = A \underline{s}_t + B \underline{a}_t$

$$\underline{s}_{t+1} = f(\underline{s}_t, \underline{a}_t)$$

$$\underline{s}_1 \xrightarrow{\underline{a}_1} \underline{s}_2 \quad \dots \quad \underline{s}_{T-1} \xrightarrow{\underline{a}_{T-1}} \underline{s}_T$$

LQR:

minimize

$\{a_t\}_{t=1}^{T-1}$

$$\left(\sum_{t=1}^{T-1} \underline{s}_t^T Q \underline{s}_t + \underline{a}_t^T R_t \underline{a}_t \right) + \underline{s}_T^T Q_T \underline{s}_T$$

s.t. $\underline{s}_{t+1} = A_t \underline{s}_t + B_t \underline{a}_t \quad \forall t=1, \dots, T-1$

Example:

$\|\underline{s}_T - \underline{s}_g\|_2^2$ ← quadratic cost

$$= (\underline{s}_T - \underline{s}_g)^T (\underline{s}_T - \underline{s}_g)$$

$$= \underbrace{\underline{s}_T^T \underline{s}_T}_{\text{quad}} - 2 \underbrace{\underline{s}_g^T \underline{s}_T}_{\text{linear}} + \underline{s}_g^T \underline{s}_g$$

$$= \underbrace{\begin{bmatrix} \underline{s}_T^T & 1 \end{bmatrix}}_{\tilde{\underline{s}}_T} \underbrace{\begin{bmatrix} I_{n \times n} & -\underline{s}_g \\ \underline{s}_g^T & \underline{s}_g^T \underline{s}_g \end{bmatrix}}_{\tilde{Q}_T} \underbrace{\begin{bmatrix} \underline{s}_T \\ 1 \end{bmatrix}}_{\tilde{\underline{s}}_T} = \tilde{\underline{s}}_T^T \tilde{Q}_T \tilde{\underline{s}}_T$$

Expressed
in the LQR
cost function
form

Trade off between
energy spent and
going to goal

$$\text{cost} = \|\underline{s}_t - \underline{s}_g\|_2^2 + \frac{1}{2} m v_t^2 + \frac{1}{2} I \omega_t^2$$

$$\underline{a}_t = \begin{bmatrix} v_t \\ \omega_t \end{bmatrix}$$

$$= \underbrace{\tilde{\mathbf{z}}_t^T}_{\mathbf{\bar{a}}_t} \mathbf{Q}_t \tilde{\mathbf{z}}_t + \underbrace{[\mathbf{v}_t \ \mathbf{w}_t]^T}_{\mathbf{R}} \underbrace{\begin{bmatrix} \frac{1}{2} \mathbf{m} & 0 \\ 0 & \frac{1}{2} \mathbf{I} \end{bmatrix}}_{\mathbf{a}_t} \begin{bmatrix} \mathbf{v}_t \\ \mathbf{w}_t \end{bmatrix}$$

$\mathbf{Q}_t$ matrix captures preferences over states

$\mathbf{R}_t$ matrix typically penalizes big actions
and leads to smoother trajectories

Unicycle system dynamics is not Linear
non-linear

$$\begin{bmatrix} x_{t+1} \\ y_{t+1} \\ \theta_{t+1} \end{bmatrix} = \begin{bmatrix} x_t + v_t \cos \theta_t \Delta t \\ y_t + v_t \sin \theta_t \Delta t \\ \theta_t + \omega_t \Delta t \end{bmatrix}$$

$$\mathbf{s}_{t+1} = \mathbf{A} \mathbf{s}_t + \mathbf{B} \mathbf{a}_t$$

which terms are non-linear?

Try to make linear

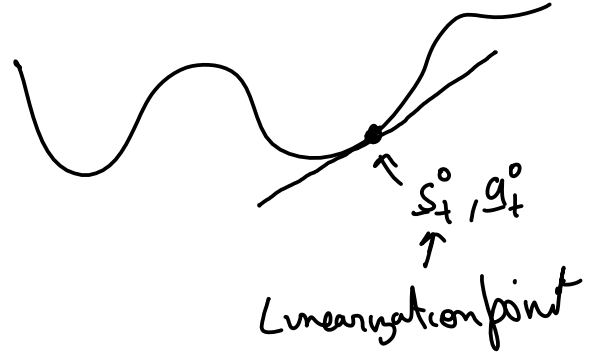
$$\begin{bmatrix} x_{t+1} \\ y_{t+1} \\ \cos \theta_{t+1} \\ \sin \theta_{t+1} \end{bmatrix} = \begin{bmatrix} x_t + v_t \cos \theta_t \Delta t \\ y_t + v_t \sin \theta_t \Delta t \\ \cos \theta_t + \text{~~~~~} \\ \sin \theta_t + \text{~~~~~} \end{bmatrix}$$

Is it Linear now?, If not due to which terms?

Taylor series is used for local linear approx.

$$\underline{s}_{t+1} = \underline{f}(\underline{s}_t, \underline{a}_t)$$

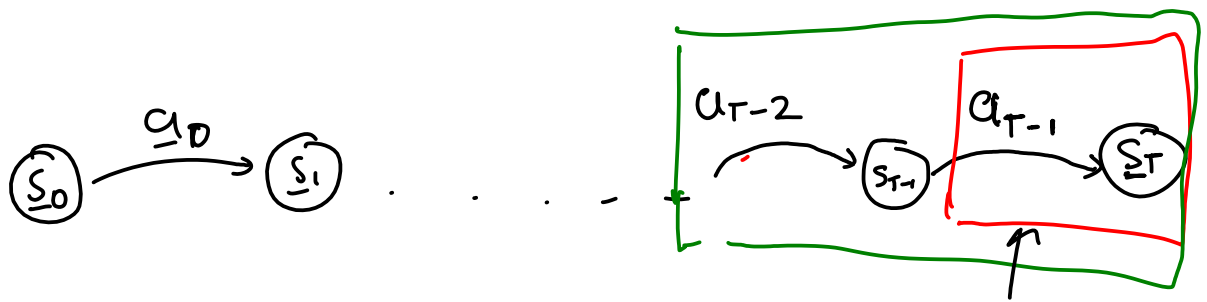
Linearize around $\underline{s}_t^o, \underline{a}_t^o$



$$\underline{f}(\underline{s}_t, \underline{a}_t) \approx \underline{f}(\underline{s}_t^o, \underline{a}_t^o) + \left[\underline{J}_{\underline{s}_t} \underline{f} \right] (\underline{s}_t - \underline{s}_t^o) + \left[\underline{J}_{\underline{a}_t} \underline{f} \right] (\underline{a}_t - \underline{a}_t^o)$$

Jacobien matrices

$$\approx \underline{f}(\underline{s}_t^o, \underline{a}_t^o) + \frac{\partial \underline{f}}{\partial \underline{s}_t} (\underline{s}_t - \underline{s}_t^o) + \frac{\partial \underline{f}}{\partial \underline{a}_t} (\underline{a}_t - \underline{a}_t^o)$$



LQR:

minimize

$\{a_t\}_{t=1}^{T-1}$

$$\left(\sum_{t=1}^{T-1} \underline{s}_t^T Q_t \underline{s}_t + \underline{a}_t^T R_t \underline{a}_t \right) + \left(\underline{s}_T^T Q_T \underline{s}_T \right)$$

s.t. $\underline{s}_{t+1} = A_t \underline{s}_t + B_t \underline{a}_t \quad \forall t=1, \dots, T-1$

minimize

a_{T-1}

$$\underline{a}_{T-1}^T R \underline{a}_{T-1} + \underline{s}_T^T Q \underline{s}_T$$

s.t.

$$\underline{s}_T = A \underline{s}_{T-1} + B \underline{a}_{T-1}$$

$$\equiv \text{minimize}_{a_{T-1}} \quad \underline{a}_{T-1}^T R \underline{a}_{T-1} + \left(A \underline{s}_{T-1} + B \underline{a}_{T-1} \right)^T Q \left(A \underline{s}_{T-1} + B \underline{a}_{T-1} \right)$$

How to minimize a Quadratic vector function?

For scalar function:

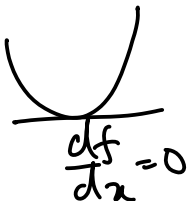
$$f(x) = ax^2 + bx + c$$

$$\frac{df}{dx} = 0$$

$$f(x) = a \left(x^2 + \frac{b}{a} x \right) + c$$

$$= a \left(x^2 + \frac{b}{a} x + \frac{b^2}{4a^2} - \frac{b^2}{4a^2} \right) + c$$

$\forall a > 0$



$$f(x) = \underbrace{a \left(x + \frac{b}{2a} \right)^2}_0 + \underbrace{c - \frac{b^2}{4a}}_{\text{if } a > 0}$$

$$f(x) \text{ minimum at } x = -\frac{b}{2a}$$

Can we do something similar for vector function

$$f(\underline{x}) = \underbrace{\underline{x}^T A \underline{x}}_{\substack{\uparrow \\ \text{vector}}} + \underbrace{\underline{b}^T \underline{x}}_{\substack{\uparrow \\ \text{matrix}}} + c$$

$$\frac{\partial}{\partial \underline{x}} \underline{x}^T A \underline{x} = 2 \underline{x}^T A$$

$$a > 0,$$

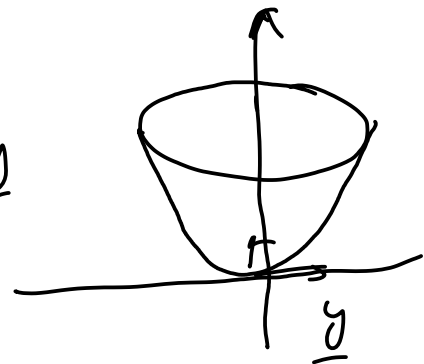
$$\frac{\partial}{\partial \underline{x}} \underline{b}^T \underline{x} = \underline{b}^T$$

$A > 0$? Positive definiteness (PD)

A matrix A is PD if $\underline{y}^T A \underline{y} > 0$ for all $\underline{y} \in \mathbb{R}$

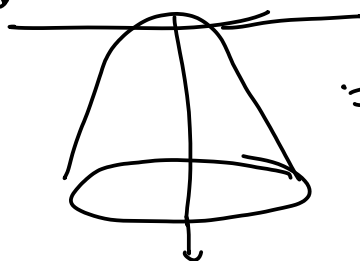
PD $\equiv$ all eigen values are +ve

$$\underline{y}^T A \underline{y}$$



negative definite

$$\underline{y}^T A \underline{y} < 0$$



$\equiv$ all eigen values are -ve

saddle point



$\equiv$ some +ve, some -ve

$$\underline{y}^T A \underline{y} = \lambda \underline{\underline{y^T y}} =$$

eigen vectors
= unit vector

$$\underline{y}^T A \underline{y} = \lambda \|\underline{y}\|^2$$

$$\frac{\underline{y}^T A \underline{y}}{\|\underline{y}\| \|\underline{y}\|} = \lambda \leftarrow \text{eigen value}$$

$\forall \underline{y}^T A \underline{y} > 0 \longrightarrow$ eigen value are +ve